

[DC] Immersive disorientation - Visualizing Dolly Zoom through Counter-Invariant Perception in Cinematic Virtual Reality

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ABSTRACT

Cinematic virtual reality (CVR) is a relatively new academic discipline within the field of VR studies. Many areas remain unexplored, notably cinematography for CVR. My doctoral research focuses on the camera movement in CVR, i.e., the dolly zoom. While the well-established technique creates visual distortion through camera movement in traditional cinema, it cannot be applied directly in CVR due to the optical limitations. Thus, the research explores the possibility of representing the perceptual distortion of the dolly zoom supported by rigorous scientific principles.

This paper consists of three parts. First, it introduces the background and research questions of the proposed research. Second, it discusses theories and the methodology outlining the project. Third, it recapitulates the work done and proposes further work to complete the research. The key focus of the research is simulating the dolly zoom effects by manipulating the 3DoF project parameters and creating an inverted scaling shader for 6DoF experiences. It will contribute to the development of CVR by introducing a new research direction in cinematography, a field that is currently understudied in scholarly research.

Index terms: Cinematic Virtual Reality, Dolly zoom, Visual Perception, Immersion, Disorientation

1 INTRODUCTION

CVR is an immersive experience with narrative works in a 360° environment with a fixed-story outcome, rather than an open-ended 'game'-like experience. [1] Thus, this paper studies CVR experiences in both 360° three-degree-of-freedom (3DoF) and the game-like six-degree-of-freedom (6DoF). Meanwhile, the dolly zoom combines a simultaneous dolly shot with a zoom shot in contrary motion, resulting in an optical distortion. [2] This technique is specifically designed for a camera mechanism that cannot be easily adopted in 3DoF or 6DoF CVR experiences.

Based on the above definition and findings, this research aims to visualize the dolly zoom effect in both 3DoF and 6DoF CVR experiences. It is framed with the following research questions.

1. How do users perceive and interpret the dolly zoom effect differently in 3DoF and 6DoF CVR experiences?
2. What is the potential impact of the CVR dolly zoom visualization beyond its perceptual effects?
3. How can the dolly zoom visualization enrich the narrative-based CVR?

2 THEORIES, METHODOLOGIES, AND OUTLINE

This research is grounded in Gibson's ecological approach to visual perception and Sobchack's phenomenological perspective on camera movement. Gibson's theory of the visual control of

locomotion suggests that the patch of arrays magnifies when the viewer approaches an object. [3] Furthermore, it references Sobchack's concept of viewing view/viewed view, which considers a film experience, including camera work, as a phenomenological experience that allows for the perception of the temporal film space. [4]

The visual effect of the dolly zoom violates Gibson's argument. With this finding, this research proposes the principle of counter-invariance, in which the object's appearance remains unchanged when the viewer approaches it. Inspired by the two key theories, this research explores the counter-invariant principle using computer graphics to visualize the dolly zoom effect in 3DoF and 6DoF experiences. For the 3DoF visualization, the study attempts to manipulate the longitude and latitude projection parameters while the camera movement occurs in a virtual environment (VE). For the 6DoF experience, the research develops a camera-based responsive scaling shader that scales the object's size inversely in relation to the viewer's location. Both studies focus on the first-person point of view experience of perceiving the counter-invariant distortion.

The research is structured in three parts. First, it creates prototypes to test the efficacy of the techniques in 3DoF and 6DoF VE. Second, it applies the methods to the staircase scene in Hitchcock's *Vertigo* as a cinematographic study in CVR. [5] Third, it utilizes the techniques to create a Ten Court of Hell 3DoF CVR experience, informing and educating the audience.

All three parts will be evaluated with semi-structured interviews on a phenomenological basis. The interview will focus on the viewer's perception and understanding of the CVR experiences, along with psychological questionnaires such as the Positive and Negative Affect Schedule (PANAS) or the Presence Questionnaire.

3 WORK IN PROGRESS

Until this stage, this research has already completed the 3DoF CVR dolly zoom visualization. The visualization involves three phases: planar stereoscopic dolly zoom, equirectangular zoom simulation, and the stereoscopic dolly zoom based on the equirectangular projection. The planar stereoscopic dolly zoom is a rendering technique for binocular displays (primarily head-mounted displays), which offers a sense of depth when the dolly zoom effect is applied. The equirectangular zoom simulation is to change the Field of View (FOV) and approximate the zoom effect in a 360° VE by remapping the longitude and latitude parameters in the equirectangular projection. The stereoscopic dolly zoom is the ultimate form of the dolly zoom visualization in the 3DoF CVR. [Fig. 1] It incorporates the mentioned techniques to simulate the counter-invariant principle in the dolly zoom. The development of the 6DoF CVR dolly zoom visualization is ongoing. Nevertheless, a camera-based responsive scaling shader has been developed and implemented in a first-person shooter game mode experience. In the experience, the object shrinks when the viewer approaches it,

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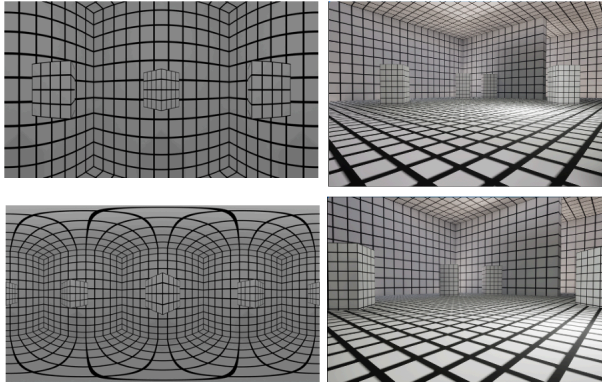


Figure 1 Dolly zoom visualization in 3DoF VE (left), and in 6DoF (right)

and vice versa, resembling the counter-invariant effect in the dolly zoom. [Fig. 1] This shader is ready to be executed in a 6DoF VR setting.

For applying the techniques to the *Vertigo* staircase scene, this research has almost completed the 3DoF CVR visualization. After various versions of testing, the CVR experience is a continuous 360° animation from the protagonist's point of view. It visualizes the chase between the protagonist and the woman, as well as the dolly zoom shot of looking down the stairs. [Fig. 2] However, further editing and post-production are needed to optimize the visual cues and rhythm of the animation. For the 6DoF visualization, a first-person perspective walkthrough prototype is ready for simulating the counter-invariant effect. The scaling shader is applied to the staircase to visualize the disorientation during the dolly zoom. (also see Figure 2) Although the perception of disorientation is successfully simulated, the cinematic narrative has yet to be translated sufficiently to the 6DoF VE. Thus, further work is needed before implementing VR.



Figure 2 CVR dolly zoom application in Vertigo's staircase: in 3DoF VE (top), in 6DoF VE (bottom)

Building on the technical and cinematic prototypes, the Ten Court of Hell CVR experience is still in the early stages of production. This experience references the demolished mural paintings depicting judgment and punishment in the Taoist hell in the Hong Kong Haw Par Mansion. As a cultural heritage project,

the objective is to reinforce the sense of discomfort in hell, which is the key message of the original mural paintings. After the background research, I have already designed the characters in the CVR scenes, including the kings, officers, bailiffs, and the protagonists. Based on the story, the CVR experience will consist of five scenes, while four of them depict the courts of hell. The first scene, Court One for judgment, has already been designed. [Fig. 3] It encompasses the key Chinese structure, the bridges for reincarnation, the underwater visual, and the corresponding character. After building all scenes and animating the characters, I will explore the camera work, especially the dolly zoom. The key function of the dolly zoom in this CVR experience is to transition between scenes and create perceptual disorientation, simulating the vertigo of underworld punishment and moral judgment. The perceptual discomfort aims to enhance the educational impact of the dismantled mural paintings.

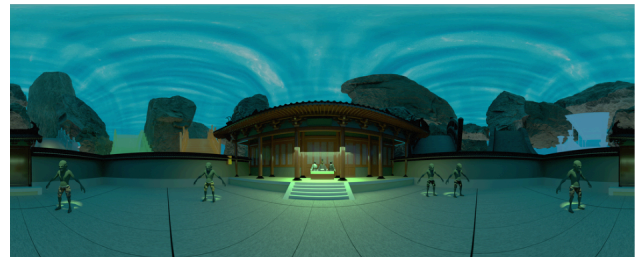


Figure 3 3DoF Ten Court of Hell Scene Setting (Court 1)

CONCLUSION

The proposed research has three key contributions. First, it introduces the counter-invariance for simulating the dolly zoom in CVR. Second, it suggests the projection manipulation for the simulation in the 3DoF VE. Third, it designs the camera-based responsive scaling shader for the 6DoF VE.

I would appreciate feedback on the evaluation design and advice on the perceptual study design of my practice. CVR Critiques are highly welcome, as they will help me position the research and strengthen its contribution to CVR.

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